

Appendix 1: Systematic Literature Review Process

Appendix 1 aims to systematically review the current state of blockchain-based design patterns. To achieve this goal, we followed the guidelines proposed by Kitchenham and Charters [1], [2], [3]. The following outlines the research methodology of the SLR employed in our study.

By using the SLR, we can identify decision model elements necessary to conduct a thorough search and select accessible literature on blockchain-based design patterns. Data extraction and analysis were then conducted to determine the current state, including distribution, typical challenges and relevant quality attributes.

1 RESEARCH QUESTION

In order to gain a comprehensive understanding of the current state and impact of design patterns for decision-making, we defined three main research questions (RQs) to drive this SLR.

- **RQ1:** What is the state-of-the-art taxonomy in terms of design patterns for blockchain-based application?
- **RQ2:** What are the mappings between quality attributes and design patterns?
- **RQ3:** How to intuitively select patterns based on their taxonomy and mappings with a decision model?

Firstly, *RQ1* attempts to understand the taxonomy of blockchain-based design patterns, including their current state, common concerns and challenges, as well as distribution. Next, *RQ2* attempts to determine the relevant quality attributes, including standardized metrics and attribute distribution. Lastly, *RQ3* examines how the assessment and selection process for design patterns is supported by a decision model that employs a taxonomy and quality attributes.

2 SEARCH STRATEGY

This section discusses how to systematically and comprehensively search and collect relevant literature. Here we conduct a rigorous literature search in accordance with the recommendations [1], [4]. To complete this process, we searched papers published in scientific literature sources up to July 2023. All the papers have been queried from a wide range of scientific literature sources, to prevent relevant studies remaining hidden, including:

- *Digital libraries:* ACM Library (113), IEEE Xplore (154), ScienceDirect (182), SpringerLink (254).
- *Citation databases:* Web of Science (41), Scopus (221).
- *Citation search engines:* DBLP (76)

And the Quasi-Gold Standard (QGS) [5] method was employed to select appropriate search terms. This method involves selecting a set of studies that must appear in the query results and constructing the query around the terms used in those papers. We chose this search strategy due to the considerable number of relevant papers, and can capture the most relevant research through a relatively rigorous process. To form the corpus of studies for the QGS method, we selected five typical studies concerning blockchain-based design patterns: [6], [7], [7], [8], [9]. Consequently, the search string for the query was constituted, where an asterisk (*) is used to match zero or more characters, and a question mark (?) is used to match a single character.

TITLE-ABSTR-KEY ((blockchain OR BLC OR "decentralized ledger" OR DLT OR "smart contract?") AND ("design pattern?" OR "blockchain pattern?" OR "blockchain-based pattern?" OR "BLC pattern?"))*

To enhance query precision, we deliberately restricted our search to papers that featured the relevant terms in their title, abstract, or keywords. The searches were carried out by four researchers and all results were double-checked. If necessary, the search string was adjusted to suit the specific query language of the database or search engine. String adjustments were agreed upon by all authors. Any disagreements on any of the identified papers were settled through in-depth analysis of the full text after mutual discussion. After completing the search process, we obtained a total of 1041 papers.

3 STUDY SELECTION

This section discusses how to select papers that contain useful information from the collection obtained in the previous step. To ensure that we selected primary studies that would provide valuable information for our research questions, we assessed and cross-checked the relevance of each study based on our inclusion and exclusion criteria, as shown in Table 1.

Here is a more specific description of the process. To begin with, the initial selection of 1041 papers was conducted from seven literature sources. A spreadsheet of search results was used to automatically deduplicate the papers, resulting in a total of 492 papers. In the second phase, applying the exclusion criteria to check the venue, paper title, and abstract, the total number of included papers declined substantially by four student researchers. Unrelated topics to SE (exclusion

TABLE 1: Inclusion and Exclusion Criteria

Inclusion Criteria	
I1	The paper proposes one or more blockchain-based patterns.
I2	The paper involves the case of existing blockchain-based patterns.
I3	The paper’s scope is generic without specific scenario restrictions.
I4	The paper includes at least a description of the context and solution of the pattern.
Exclusion Criteria	
E1	The paper lies outside software engineering and blockchain.
E2	The full text is not accessible.
E3	The paper is not written in English.
E4	The paper has not been peer-reviewed.

criteria 1) was the leading cause of the decline, and 214 papers were removed. We also removed 116 papers that were grey publications or were not peer reviewed (exclusion criteria 4). 65 papers that are not written in English or not available in full text (exclusion criteria 2, 3). And the results are double-checked by our researchers. In the third phase, the study’s full-text was further checked based on the inclusion criteria. At a minimum, the selected studies need to include the discussion of design patterns and practical cases such as sample codes or realistic applications. Thus 57 papers were removed (inclusion criteria 1, 2). Also, 10 papers were removed due to the fact that they provide design patterns with specific scenarios, or that their description does not include at least the context and solution (inclusion criteria 3 and 4). Finally, we introduced the snowballing approach to mitigate the risk of missing important studies [10]. 30 candidate papers were initially selected for snowballing, and 4 additional papers were added during the snowballing process. These four highly valuable articles that are closely related to blockchain design patterns. Although these articles were published on arXiv and IEEE Access without peer-review, their significance and influence in the field prompted us to include them in the scope of the SLR.

In summary, with rigorous method and process, we obtained 34 relevant studies from 1041 papers as shown in Fig. 1. The entire search and selection process was completed within nearly four weeks.

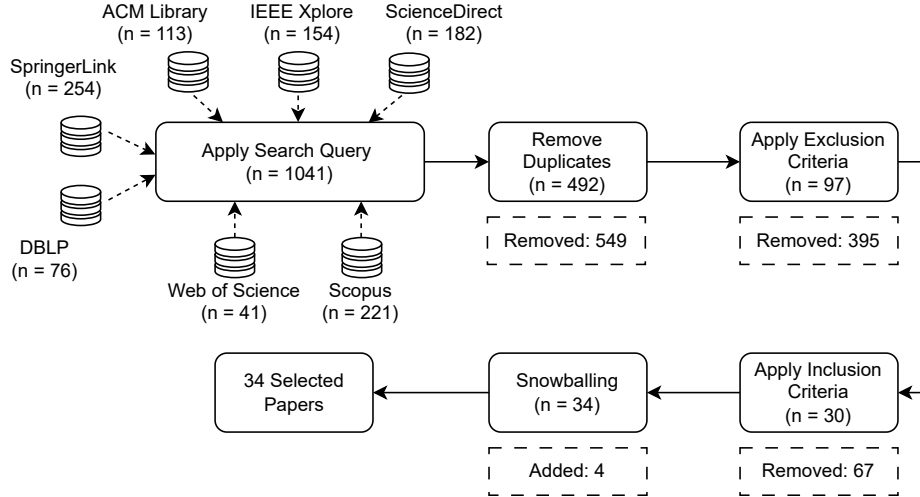


Fig. 1: Search and Selection Process of Relevant Literature

4 DATA EXTRACTION

This section discusses how to extract relevant information about design patterns from the selected papers. The data extraction process was conducted using Table 2. The first four items are metadata items, while the remaining data items are briefly explained in the right column. The data items related to the research questions are labeled RQ1 and RQ2. Since the data here is the source data for taxonomy establishment (RQ1) and mapping relationships between design patterns and quality attributes (RQ2). The decision model (RQ3) is constructed based on the results of taxonomy and mapping, and is not included in this part.

The process of data extraction was carried out by thoroughly examining the full-text of the studies. First, a pilot research was conducted before formal data extraction to ensure a standardized approach among the researchers. In our pilot study, we specifically chose five representative studies that epitomized the range and depth of patterns typically discussed in the field. These studies were selected based on their influence, citation count, and their exemplary description and demonstration of the blockchain-based design patterns. By focusing on such archetypal sources, we aimed to comprehensively test and refine our data extraction template. The pilot extraction results were cross-checked to guarantee a consistent

TABLE 2: Data Extraction Form

ID	Title	RQ	Comments
1	Authors	Metadata -	
2	Year	Metadata -	
3	Venue Type	Metadata	Conference, Journal, Workshop
4	Identifier	Metadata	Digital object unique identifier (DOI) of paper
5	Pattern Name	RQ1	Name of the pattern used in the literature
6	Pattern Context	RQ1	Design situation giving rise to a design problem [11].
7	Pattern Solution	RQ1	A form or rule that can be applied to resolve these problems [11].
8	Pattern Benefit	RQ2	Positive impacts when applying the pattern
9	Pattern Drawback	RQ2	Negative impacts when applying the pattern
10	Pattern Application	RQ2	Application cases of design patterns

understanding of the data items. Any inconsistencies in the findings were resolved through regular meetings aimed at reaching a consensus. Following this, a refined review protocol was implemented, which reflected the group’s collective agreement about the design pattern.

Second, after pilot research, the remaining studies were assigned for review. The authors thoroughly read the full-text of all identified studies and meticulously recorded the relevant data in spreadsheets according to the refined protocol. In certain instances, extracting the necessary information for comprehending the patterns presented in the reviewed papers posed a challenge, as a result of inadequate or unclear descriptions of the pattern context or solution approach. To overcome such challenges, supplementary resources such as follow-up papers and online sources were consulted for assistance. In conclusion, the data was reviewed and extracted based on all the available information.

5 QUALITY ASSESSMENT

This section discusses how to assess whether a design pattern’s description in the literature is sufficiently detailed and valuable. To analyze the overall quality of each pattern and the comprehensiveness of the pattern descriptions, a checklist comprising four questions (or quality criteria) is developed. The quality of the extracted patterns is evaluated by utilizing the pattern criteria, in conjunction with the template of design patterns as defined in relevant literature [11], [12], [13]. As shown below:

- Q1: Does the pattern have a clear description of the context, solution and problem?
- Q2: Does the pattern refer to any specific practices or code examples?
- Q3: Is the pattern presented in a standard format (e.g. GoF/Alexandrian templates [14], [15])?
- Q4: Are the pattern consequences analysed based on the pattern solution or application case?

In the checklist consisting of four questions, each question had three possible answers: Yes, Partially, and No. By determining the answers, the overall quality of the pattern could be assessed. And we only included patterns where the answer to Q1 and Q2 was at least “Partially”. This decision was made because it is challenging to extract a clear pattern without a description of the problem and the solution in a specific context. Additionally, it is difficult to assess the impact of pattern applications without specific cases or examples as supplementary information or evidence. And our analysis can be more accurate and effective if the discussion of patterns in the source literature satisfies Q3 and Q4. Based on the assessment criteria, some design patterns are excluded after discussion and consensus, and the reasons for exclusion are clarified.

6 DATA ANALYSIS

In this section, after quality assessment, the collated data is summarized and analyzed in a manner suitable for establishing the decision model. This includes the analysis of design pattern taxonomy based on the blockchain hierarchy and analysis of quality attributes describing pattern consequences.

Regarding the analysis of design pattern, during the analysis of design patterns, we initially identified 179 distinct patterns from the literature. However, we observed that many studies used different names to refer to the same pattern. To overcome this inconsistency and simplify the decision-making process, we examined each pattern and identified the same solution with different names from various sources. Therefore, after merging duplicate patterns, we finally arrived at 72 unique design patterns relevant to blockchain application development.

To systematically organize the 72 design patterns for building a decision model, an appropriate and effective classification is needed. Here we used a methodology for constructing the taxonomy, originally proposed by Nickerson et al. [16]

and further developed by D. Kundisch et al. [17]. Taxonomy was created to build a concise, robust, and extensible set of dimensions. Each dimension comprises a set of characteristics that sufficiently describe objects in a specific domain [16]. Based on the previous result, we constructed a blockchain-based design pattern taxonomy for modeling the decision model. In this study, constructing the taxonomy involves three main steps:

Firstly, based on the blockchain hierarchy, we classified the patterns according to their focus. The classification is based on the objects design patterns operate on or concentrate on. For example, patterns that operate on smart contract data or transaction data are categorized in the data layer. In contrast, patterns that focus on the smart contract code itself or on inter-invocation relationships are categorized in the contract layer. Since each design pattern has a clear focus, it can be classified reasonably. Then we identified 12 patterns for the application layer, 34 patterns for the contract layer, 2 patterns for the incentive layer, and 24 patterns for the data layer.

Secondly, to identify the concern aspects in each layer, common features need to be identified and patterns need to be further grouped. To achieve this, we introduced a Natural Language-based tool¹ for word frequency analysis. The summaries of patterns in each layer were thoroughly ingested and lemmatized, and a set of frequently recurring words was effectively identified. Combined with our empirical knowledge, these recurring words were then organized into a total of 13 relevant concern aspects in relation to the patterns.

Finally, in order to further conclude the typical challenges in each aspect, we conducted a similar iteration process for the context and problem of the design pattern. Through this iteration, combined with our understanding, we identified 28 different types of challenges. For a design pattern that addresses multiple challenges, we focus on its primary aspect and its original intention. Some patterns' mitigation effect on other challenges is expressed through the associative relationship between the challenges. The resulting taxonomy can be found in our paper, which is associated with RQ1-a.

Additionally, we conducted a *quantitative analysis of blockchain patterns* to determine the popularity of the concern aspects we identified above. To understand the distribution of design patterns, we utilized the completed design pattern taxonomy as our fundamental data source, and performed further statistical analysis. For each layer and aspect, we analysed two key metrics: the number of literature cited and the number of patterns mentioned. The former metric reflects the breadth of discussion on a given aspect/layer, while the latter metric indicates the depth (level of attention) paid to a given aspect/layer. By repeating the statistical operations for each layer, we can analyze the distribution patterns of design patterns within the blockchain hierarchy. Overall, through quantitative analysis, we sought to identify which aspects are most commonly focused when proposing design patterns, which can be found in our paper.

Regarding the analysis of quality attributes, in order to assess design patterns and their consequences, we propose the metrics of quality attributes based on the SLR. In addition to the literature, the impact of a pattern may differ across practical cases. To complement our previous research based on the literature, we examined specific application cases or examples of the extracted design patterns, and summarized their effects. Thus, all 18 quality attributes have been derived from existing studies or practical applications.

Due to the numerous design patterns, a unified standard is needed to evaluate their impact through quality attributes. Therefore, we require a baseline for consistency of quality attributes among patterns, which can be achieved by comparing with a baseline blockchain system. Since the proposed decision model is a generic model, the baseline blockchain is also an abstract concept with only typical generic attributes such as decentralization, transparency, etc.

Based on the pattern description and the practical case, it is relatively easy to determine whether a pattern has positive or negative effects on quality attributes. The applicability of the quality attributes in different contexts is more challenging to decide. And we divide the quality attribute contexts (or scopes) into three categories: *Blockchain generic attributes* are the mapped characteristics reflected in all blockchain systems, which make up the majority. *Pattern-specific attributes* are the mapped characteristics presented in the blockchain system applying to particular patterns. In addition, some of the patterns are optimizations for traditional systems, so we added a category: *Traditional system attributes* to indicate the mapped characteristics compared with traditional systems. Finally, considering the definition in ISO [18] and IEEE [19], [20], as well as the features of blockchain-based applications, we have identified 18 distinct and highly relevant quality attributes and defined a unified standard to assess the impact of each pattern.

According to the standard we proposed, we therefore analyzed the consequences of the design patterns, including positive and negative impacts, and represented them with these attributes. In order to construct clear decision paths, we also analyzed the commonalities and differences among the quality attributes. All relevant information about quality attributes is presented in the decision model, associated with RQ2. Further details can be found in our paper.

We also conducted a *quantitative analysis of quality attributes* to determine the degree of attention. Building on the previously identified relationship between design patterns and quality attributes, we conducted a frequency analysis of each attribute mentioned. And we identified 254 instances where quality attributes were mentioned. These were then categorized into positive and negative attributes and further analyzed in different layers. Statistical charts were employed to analyze the tendency of quality attributes at different layers. As a result, we were able to provide a distribution of quality attributes across different layers. Further details of the results and analysis are discussed in our paper.

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